

Advanced Radio License Notes

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1 Introduction

This are the notes from following the advanced course on HamRadio.au.

2 Electronics and Circuit Theory

2.1 Impedance

2.1.1 Introduction

Impedance is how a circuit stores or resists AC (or any changing) energy. If the impedances within a circuit aren't matched, then power gets reflected back instead of going where it is needed.

Impedance is the combination of resistance and reactance. Resistance is the restriction of the "flow" of energy, and converts that restricted energy to heat. This is measured in Ohms Ω . Reactance can be thought of as a flywheel, it stores energy temporarily and releases it. This is what capacitors and inductors do. This is also measured in Ohms. This means that impedance is the total opposition to current.

Why this matters. The transmitter expects to see 50Ω of impedance. The coax is also 50Ω . If the antenna isn't also close to 50Ω , then the power from the transmitter gets reflected, which is what SWR (standard wave ratio) measures.

2.1.2 Reactance

There are two types of reactance. One is found in inductors, and the other in capacitors.

Inductors (any coil of wire) opposes changes in current, i.e. their reactance increases with frequency

$$X_L = 2\pi fL.$$

This means that the higher the frequency, the more an inductor blocks the signal. This is why they are used in low-pass filters, they all for low frequencies but block high ones.

Capacitors oppose changes in voltage. Their reactance decreases with frequency

$$X_C = \frac{1}{2\pi fC}.$$

The higher the frequency, the easier it passes through the capacitors. This is why a capacitor in series blocks DC but passes RF.

2.1.3 Impedance formula

We can then write impedance as

$$Z = R + jX,$$

where R is the resistance, X is the impedance, and j is used in place of i for imaginary numbers due to i being used for current. The resistance part is where the energy is consumed, and the reactive part is where the energy is stored and returned. The total impedance is then given by the magnitude of this complex number

$$|Z| = \sqrt{R^2 + X^2}$$

Example. An antenna has $R = 50\Omega$ and $X = 25\Omega$. The total impedance magnitude is

$$\begin{aligned} Z &= \sqrt{2500 + 625} \\ &= \sqrt{3125} \\ &\approx 56\Omega \end{aligned}$$

2.2 Resonance

2.2.1 Introduction

When an inductor and capacitor are combined, at one specific frequency their reactances become equal and therefore cancel each other out. This is called resonance, and when this happens the circuit behaves as purely resistive. This is how a single frequency is selected while rejecting everything else.

Radio Example. Tuned circuits within a receiver use this resonance to select a station to hear, and reject everything else. Antennas are resonant at a specific frequency, which is where it works best.

The formula to find the resonant frequency is given by

$$f_r = \frac{1}{2\pi\sqrt{LC}}.$$

This says that for a given inductor L and capacitor C , there's exactly one frequency where they resonate. This is an important formula.

2.2.2 Series and Parallel Resonance

Series and parallel arrangements of L and C result in opposite behaviour at resonance.

Table 1: Behaviour of circuits at resonance		
	Series LC	Parallel LC
Impedance	Minimum (just R)	Maximum
Current	Maximum	Minimum from source
Used for	Passing a desired frequency (band-pass)	Blocking a desired frequency (or selecting it) (band-stop)

Radio Example. A series-resonant circuit in an antenna trap passed the trap's frequency through, while a parallel-resonant circuit in a receiver's IF (intermediate frequency) stage has high-impedance at the desired frequency, which develops a large signal voltage across it.

2.2.3 Q-Factor

The Q-factor describes how selective a resonant circuit is. A higher Q means a very sharp tuning (with narrow bandwidth), and low Q means a very broad tuning (wide bandwidth).

$$Q = \frac{f_r}{\text{Bandwidth}}$$

For a series RLC , $Q = \frac{X_L}{R}$, so less resistance means higher Q.

Bandwidth Rule. Bandwidth is $\frac{f_r}{Q}$, so a circuit with $Q = 100$ at 7 MHz has a bandwidth of 70 kHz, which is enough to cover a portion of the 40m band. A crystal filter with $Q = 50000$ at 9 MHz has a bandwidth of only 180 Hz, which is perfect for CW.

2.3 Filters

2.3.1 Introduction

Filters are extremely important in radio. They are what selects the frequencies that get let through and what get blocked. Without filters, a receiver would hear literally everything at the same time, and the transmitter would spray harmonics all over the spectrum (which happens to be illegal).

2.3.2 Types of Filter

There are four basic types of filter.

2.3.3 LC Filter

Filters use inductors and capacitors in combinations to make a specific filter. From above, we know

- Inductors block high frequencies (reactance increases with frequency)
- Capacitors block low frequencies (reactance decreases with frequency)

Table 2: Basic types of filter

Filter Type	Behaviour	Where it is in radios
Low-pass	Passes low frequencies, blocks high frequencies	After the transmitter PA (power amplifier). It blocks harmonics before reaching the antenna
High-pass	Passes high frequencies, block low frequencies	At a TV antenna input. It blocks HF signals, but lets VHF/UHF TV through
Band-pass	Passes a specific range, blocks everything else	IF (intermediate frequency) filters in a receiver. It selects the signal bandwidth that you want.
Band-stop (notch)	Blocks a specific range, passes everything else	Notch filter to remove an interfering carrier.

Low-pass Filter. A low-pass filter wants to block high frequencies, and allow low frequencies through. This means that the circuit would put an inductor in series (to block high frequencies), and capacitors to ground (short high frequencies to ground).

2.3.4 Filter Steepness

In practice, filters can't instantly go from passing a frequency to blocking another. The change is gradual, and this is affected by the number of LC sections (called a pole). Each pole adds 20 dB/decade of rolloff. The higher this number, the more steep the dropoff, as it is saying that per decade (1 decade is a ratio of 10 between two numbers on a logarithmic scale), the scale drops by 20 dB.

Example. Consider a transmitter that produces a 7 MHz signal. The 2nd harmonic at 14 MHz and 3rd harmonic at 21 MHz must be suppressed. A good 5-element low-pass filter with a cutoff just about 7 MHz can suppress the 2nd harmonic by 40+ dB and the 3rd by 60+ dB. This is the difference between a clean signal and an interference complaint.

A typical transmitter low-pass filter has 5-poles (100 dB/decade, a steep slope). For critical applications, a 7-pole filter is used (140 dB/decade, a very steep slope).

2.3.5 Crystal Filters

Crystal filters have very high Q factors (10 000 to 100 000+). Multiple crystals combined in a so-called ladder filter give the sharp selectivity for IF stages

- 2.4 kHz bandwidth - for SSB reception
- 500 Hz bandwidth - for CW reception
- 6 kHz bandwidth - for AM reception

In modern radios however, crystals are being replaced with DSP (digital signal processing) filters, as they allow changing the bandwidth on the fly.

2.3.6 Filter Response Shapes

Different mathematical designs give different trade-offs. Butterworth designs have a smooth, flat passband with no ripple. This is used where consistent gain across the passband matters. Chebyshev has a steeper rolloff than Butterworth, but some ripple in the passband. This is used where sharp selectivity matters more than a perfectly flat response.

2.4 Amplifier Classes

2.4.1 Introduction

The final stage of a transmitter is a power amplifier. This amplifier will have some bias, and depending on how that bias is configured changes two important things. First is the linearity - how faithfully does the amplifier reproduce the signal shape (critical for SSB and AM). The second is efficiency - how much of the DC power becomes RF (all excess is heat).

2.4.2 Classes

These are the different classes of amplifiers

Table 3: Amplifier classes

Class	How it works	Efficiency	Linearity	Used for
A	Transistor conducts all the time (360°). Always on	25 – 50%	Excellent	Low-power stages, driver stages
AB	Conducts most of the time (180°-360°). Slightly biased below full-on	50 – 65%	Very good	Standard choice for SSB power amplifiers
B	Conducts exactly half the time (180°). Needs push-pull pairs	65 – 78%	OK	Push-pull audio amps
C	Conducts less than half the time (<180°). Very "punchy"	75 – 85%	Poor	FM and CW only

Important. Class C amplifiers are not suitable for SSB or AM. Both of these carry information in the variation in amplitude, so when class C clips the signal information is destroyed. It is fine for FM and CW as these modes have a constant amplitude.

Most HF transceivers use Class AB for the final PA. It is linear enough to handle SSB without excessive distortion, and efficient enough that a 400W output doesn't require a large heatsink. If a Class AB amp is overdriven (i.e. mic gain is too high), it starts producing intermodulation distortion. This is a "splatter" that interferes with adjacent frequencies.

2.4.3 Decibels

Decibels allows for working with ratios by simply adding and subtracting instead of multiplying and dividing.

Table 4: Shortcuts for working with decibels

dB	Power ratio	Voltage ratio	Notes
+3 dB	$\times 2$	$\times 1.41$	Double the power
+6 dB	$\times 4$	$\times 2$	Double the voltage
+10 dB	$\times 10$	$\times 3.16$	Ten times the power
+20 dB	$\times 100$	$\times 10$	Ten times the voltage
-3 dB	$\times 0.5$	$\times 0.71$	Half the power

Example. An amplifier has 13 dB of gain. That is $10 + 3 = \times 10 \times \times 2 = \times 20$ power. Put in 20W, get out 400W.

Info. The factor is 10 for power, 20 for voltage. That is because power is proportional to the voltage squared.

Power formula: $G_{dB} = 10 \log_{10}(P_{out}/P_{in})$.

Voltage formula: $G_{\text{dB}} = 20 \log_{10}(P_{\text{out}}/P_{\text{in}})$.

2.5 Oscillators

2.5.1 Introduction

An oscillator generates a continuous AC signal at a specific frequency. This is what the transmitter uses to generate the carrier signal, and the receiver needs one to convert signals to the IF. An oscillator is an amplifier where some of the output is fed back into the input in-phase (positive feedback).

In order for this to work correctly, two criteria must be met (known as the Barkhausen criteria). The first is that the loop gain must be greater than or equal to 1. The fed-back signal must be at least as strong as what is needed at the input. The second criteria is that the fed-back signal must be in phase so that it can reinforce the original signal.

2.5.2 Common Types

Colpitts Oscillator: uses two capacitors and one inductor. The two capacitors form a voltage divider for the feedback. It is very popular in VFOs (Variable Frequency Oscillators) as it is stable and easy to tune by varying one of the capacitors.

$$f = \frac{1}{2\pi\sqrt{L \cdot \frac{C_1 C_2}{C_1 + C_2}}}$$

Hartley Oscillator: uses one capacitor and a tapped inductor (or two inductors). The tapped inductor provides the feedback. This is less common in modern equipment.

Crystal Oscillator: uses a quartz crystal as the frequency determining element. The crystal has an extremely high Q (10 000+), making it incredibly stable. It is used for reference oscillators in synthesisers, BFO (Beat Frequency Oscillator) in CW/SSB receivers, and any application where frequency accuracy is critical.

Radio Connection. When tuning a HF radio, usually you are not changing the oscillator directly, rather modern radios use a PLL (Phase-Locked Loop) synthesiser. This is a crystal oscillator that provides the stable reference, and the PLL multiplies it to the frequency that you want.

2.5.3 PLL Frequency Synthesiser

The PLL is how modern radios generate precise, tunable frequencies. In simple terms:

1. A crystal reference provides an accurate, stable signal (e.g. 10 kHz)
2. A VCO (Voltage-Controlled Oscillator) generates the output frequency
3. A divider divides the VCO output down
4. A phase detector compares the divided output with the reference
5. If they don't match, it adjusts the VCO until they do.

$$f_{\text{out}} = N \times f_{\text{ref}}$$

Example. Consider a reference $f_{\text{ref}} = 10$ kHz and $N = 1450$. The output is then $f_{\text{out}} = 14.500$ MHz (20 metres). Changing N to 1451 gives $f_{\text{ref}} = 14.510$ MHz. This is how a tuning knob works in 10 kHz steps.

2.5.4 Mixers

A mixer takes two signals and produces their sum and difference frequencies

$$f_{\text{out}} = f_1 + f_2 \quad \text{and} \quad f_{\text{out}} = f_1 - f_2$$